



PHYSICS HSSC-I SECTION – A (Marks 17)

Time allowed: 25 Minutes

Section – A is compulsory. All parts of this section are to be answered on this page and handed over to the Centre Superintendent.

Deleting/overwriting is not allowed.

Do not use lead pencil.

حصہ اول لازمی ہے۔ اس کے جوابات اسی صفحہ پر دے کر نام مرکز کے حوالے کریں۔ کات کر دہا۔
لکھنے کی اہلیت نہیں ہے۔ سیاہ پینسل کا استعمال ممنوع ہے۔

Version No.				
3	0	0	4	1

ROLL NUMBER						

0	0	0	0	0
1	1	1	1	1
2	2	2	2	2
3	3	3	3	3
4	4	4	4	4
5	5	5	5	5
6	6	6	6	6
7	7	7	7	7
8	8	8	8	8
9	9	9	9	9

0	0	0	0	0	0	0
1	1	1	1	1	1	1
2	2	2	2	2	2	2
3	3	3	3	3	3	3
4	4	4	4	4	4	4
5	5	5	5	5	5	5
6	6	6	6	6	6	6
7	7	7	7	7	7	7
8	8	8	8	8	8	8
9	9	9	9	9	9	9

Answer Sheet No. _____

Invigilator Sign. _____

Fill the relevant bubble against each question according to curriculum:

Candidate Sign. _____

Question	A	B	C	D	A	B	C	D
1. The dimensions $[ML^2T^{-2}]$ show:	Power	Work	Velocity	Acceleration	☐	☐	☐	☐
2. If $r = 2.25 \pm 0.01 \text{ cm}$ then percentage uncertainty in " r " will be:	0.2%	0.3%	0.4%	0.5%	☐	☐	☐	☐
3. $\hat{j} \cdot (\hat{i} \times \hat{k}) =$	0	-1	1	2	☐	☐	☐	☐
4. Magnitude of centripetal force $F_c =$	$m r \omega^2$	$m r \omega$	$n \omega^2$	$m v^2$	☐	☐	☐	☐
5. If the length of blade of a fan is increased, then its rotational speed (at same voltage and current):	Will increase	Will decrease	Will become infinite	Remains constant	☐	☐	☐	☐
6. Two bodies A and B of same masses have velocities in the ratio 2:1, their kinetic energies are in the ratio:	2:1	1:2	4:1	1:4	☐	☐	☐	☐
7. Maximum drag force on a 1kg falling sphere is:	9.8 N	1 N	98 N	4.9 N	☐	☐	☐	☐
8. Young's Modulus, $Y =$	$\frac{\Delta L / L}{F / A}$	$\frac{F A}{\Delta L L}$	$\frac{\Delta L L}{F A}$	$\frac{F / A}{\Delta L / L}$	☐	☐	☐	☐
9. In an adiabatic process which quantity remains constant?	Internal energy	Volume	Temperature	Mass	☐	☐	☐	☐
10. Location of a submarine in deep sea can be detected by:	Doppler effect	Compton's effect	Faraday's Law	Bernoulli principle	☐	☐	☐	☐
11. An electron is placed in a uniform electric field directed from south to north. In which direction will electron move after it is released?	East	South	North	West	☐	☐	☐	☐
12. In order to achieve high accuracy, the slide wire of a potentiometer should be:	As long as possible	As short as possible	Very thick	Neither too small, nor too large	☐	☐	☐	☐
13. Two closely placed parallel wires are carrying current in opposite direction, the wires will:	Repel each other	Attract each other	Not affect each other	Rotate each other	☐	☐	☐	☐
14. One weber is equal to:	$N A^{-1}$	$N m A^{-1}$	$N m^{-1} A$	$N m^{-1} A^{-1}$	☐	☐	☐	☐
15. When an object moves with a very high speed its length _____ in the direction of its motion.	Contracts	Expands	Remains same	Becomes zero	☐	☐	☐	☐
16. Relativistic mass is always:	Less than rest mass	Greater than rest mass	Equal to rest mass	Zero	☐	☐	☐	☐
17. A hadron may either be a Baryon or a:	Lepton	Gluon	Photon	Meson	☐	☐	☐	☐

— 1HA-I 25004 (B) —

$$\vec{A} \cdot \vec{B} = AB \cos \theta \quad L = I \omega \quad K.E = \frac{1}{2} m v^2 \quad W = mg \quad g = 9.8 \text{ ms}^{-2} \quad v = r \omega \quad \vec{A} \times \vec{B} = AB \sin \theta \quad \phi_B = \vec{B} \cdot \vec{A} \quad \text{uncertainty} = \frac{\text{Absolute uncertainty}}{\text{No. of vibrations}}$$



PHYSICS HSSC-I

20

Time allowed: 2:35 Hours

Total Marks Sections B and C: 68

SECTION – B (Marks 42)

Q. 2 Answers the following parts briefly.

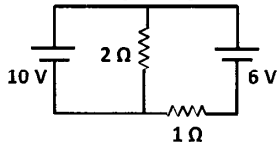
(14 × 3 = 42)

(i)	Briefly explain 'precision' and 'accuracy' with a diagram.	1x3	OR	How can ordinary light be polarized using polarizers?	03
(ii)	The velocity of a missile traveling at 1500 ms^{-1} increases to 5000 ms^{-1} after moving through a distance of 70 km. Calculate the time for the missile to reach this velocity.	03	OR	A body at 200°C undergoes a reversible isothermal process. The heat energy removed in this process is 7300 J. Calculate the entropy change and discuss entropy sign.	03
(iii)	What is meant by 'conservative' and Non-conservative' forces? Give one example each.	03	OR	What is meant by Magnus effect? Explain with the help of an example.	03
(iv)	Differentiate between crystalline solids and amorphous solids. (Give at least three differences.)	03	OR	Earth completes its one rotation about its axis in 24 hours. Calculate angular velocity of earth about its axis in radian per second.	1+2
(v)	The Doppler's effect is used in radar system. Briefly explain the working of radar system.	03	OR	Verify that: $(\vec{A} \cdot \vec{B})^2 + (\vec{A} \times \vec{B})^2 = (AB)^2$	03
(vi)	Calculate KE in electron volt and joules gained by an electron moving through a potential difference of 2 volts.	03	OR	What happens to the density of an object as it moves in a direction with very high speed relative to speed of light? Justify.	03
(vii)	Why does the resistance of LDR decrease with increase in intensity of light on it?	03	OR	State Faraday's law of electromagnetic induction. Give its mathematical form.	2+1
(viii)	Does the induced emf in a circuit depend on the resistance of the circuit? Does the induced current depend on the resistance of the circuit? Justify.	1.5 + 1.5	OR	At what speed would the mass of proton be doubled? The rest mass of proton is $1.673 \times 10^{-27} \text{ kg}$.	03
(ix)	The Kinetic energy of a charged particle moving in a uniform magnetic field does not change. Why? Explain briefly.	03	OR	If $^{233}_{92}\text{U}$ decays twice by alpha particle emission, what is the resulting isotope? Show by equation.	03
(x)	What is Faraday's cage? Briefly explain its working.	03	OR	Compare gluons and photons.	03
(xi)	What is meant by annihilation? Give one annihilation reaction.	03	OR	To increase the intensity of wave by a factor of 10, calculate the required increase in amplitude of wave.	03
(xii)	How will the radius of a flexible ring change if it is given negative charge?	03	OR	What is the difference between inertial and non-inertial frame of reference?	03
(xiii)	Use dimensional analysis to prove the equation $P = \rho gh$ is dimensionally homogeneous.	03	OR	Write three statements of second law of thermodynamic. (one in terms of entropy)	03
(xiv)	Derive the relation between torque, moment of inertia and angular acceleration.	1x3	OR	What is meant by interference? Give conditions for interference.	1+2

SECTION – C (Marks 26)

Attempt the following questions.

(Use of graph paper is not necessary. Candidates can make their own grid on answer book if required)

Q.3	State and explain vector product with an example. Also show that $\vec{A} \times \vec{B} \neq \vec{B} \times \vec{A}$	3+4	OR	What is meant by projectile? Derive the mathematical expressions of following for projectile motion: i) Time of flight ii) Range and maximum range of projectile	1+2 +4
Q.4	Explain the working of refrigerator. Also explain the coefficient of performance of refrigerator.	4+2	OR	State work-energy theorem. Explain this theorem for resistive medium.	06
Q.5	State the equation of continuity and derive its mathematical form. Or: what basis is the equation of continuity derived?	1+5 +1	OR	What is meant by elastic potential energy? Find a relation for the elastic potential energy of a material.	1+6
Q.6	Why does an astronaut feel weightless in space? How can artificial gravity be produced in a satellite? For a satellite derive expression for angular velocity, time period and frequency of satellite.	1+1 +4	OR	Find the current flowing through each resistor of the circuit. 	06

— 1HA-I 25004 (B) —

$$\begin{aligned}
 & \sin^2 \theta + \cos^2 \theta = 1 & \omega = \frac{\theta}{t} & S = \frac{\Delta Q}{T} & 1 \text{ eV} = 1.6 \times 10^{-19} \text{ J} & \vec{A} \cdot \vec{B} = AB \cos \theta & \sum \varepsilon = 0 & \omega = \frac{2\pi}{T} \\
 & \vec{A} \times \vec{B} = AB \sin \theta & E = qv & I = \frac{\varepsilon}{R} & v_f = v_i + at & v_f^2 - v_i^2 = 2aS & F_c = \frac{mv^2}{r} & \varepsilon = \frac{\Delta \phi}{\Delta t} \\
 & I \propto A^2 & \sum I = 0 & m = \frac{m_0}{\sqrt{1 - \frac{v^2}{c^2}}}
 \end{aligned}$$